

Lab Assessment Rubrics of Digital System Design (CPE-344)

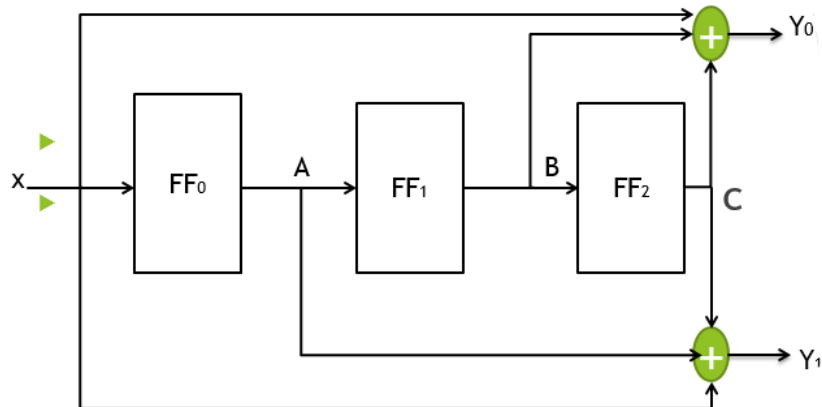
Lab Number		12								
Lab Title		Finite state machine								
Submitted by										
Names		Registration Number								
Sumaira Safeer		FA22-BCE-019								
Rubrics name & number							Marks			
							In-Lab	Post-Lab		
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.									
	R3: Interpretation of Subject Knowledge: Ability to interpret and explain mathematical and/or visual forms, including equations, diagrams, graphics, figures, and tables.									
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.									
Design	R7: Implementing Design Strategy: Ability to execute a solution taking into consideration design requirements and pertinent contextual elements. [Block Diagram/Flow chart/Circuit Diagram]									
	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.									
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.									
	R11: Tools Evaluation: Ability to simulate the experiment and them using hardware tools to verify The results.									
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.									
	R13: Management of Teamwork: Ability to appreciate, understand and work with multidisciplinary team members.									
Rubrics to follow										
Rubrics	R2	R3	R5	R6	R7	R8	R9	R11	R12	R13
In-Lab										
Post-Lab										

Lab: 12

1. Lab Tasks:

1.1. In Lab Task#1

Implement the Given Encoder in Verilog HDL



Solution:

- **Verilog Code**

```
module InLab_01(X,Clk,Rst,Y0,Y1);
input X,Clk,Rst;
output reg Y0,Y1;
reg a,b,c;
always @ (*)
begin
    Y0=X^b^c;
    Y1=X^a^c;
end
always @ (posedge Clk)
begin
    if(Rst)
        begin
            a<=0;b<=0;c<=0;
        end
    else
        begin
            a<=X;
            b<=a;
            c<=b;
        end
    end
end
endmodule
```

- **Verilog Test Bench**

```
initial begin
```

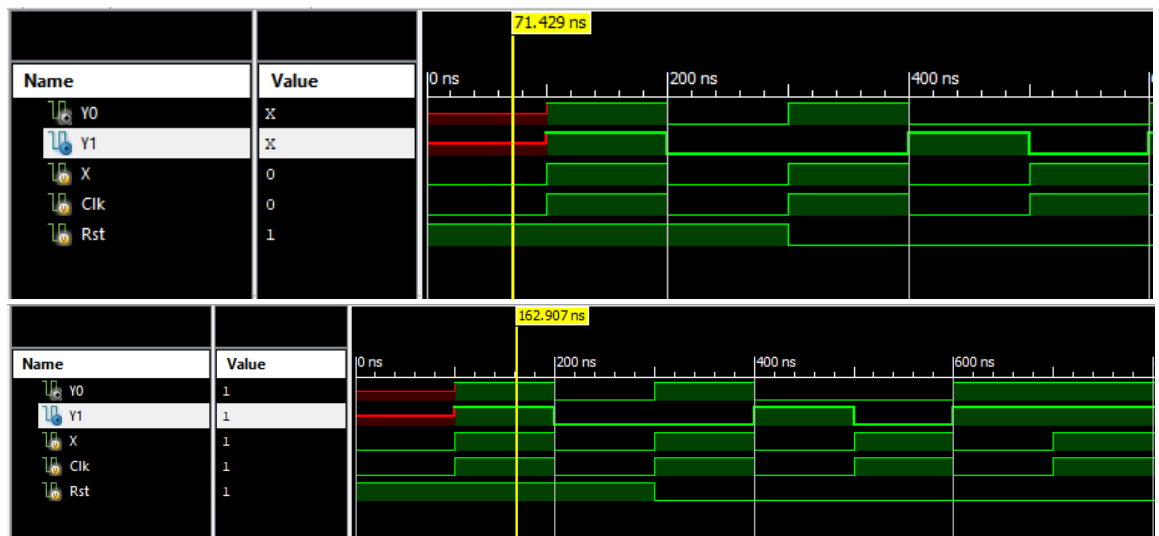
```

// Initialize Inputs
X = 0; Clk = 0; Rst = 1;#100;
X = 1; Clk = 1; Rst = 1;#100;
X = 0; Clk = 0; Rst = 1;#100;
X = 1; Clk = 1; Rst = 0;#100;
X = 0; Clk = 0; Rst = 0;#100;
X = 1; Clk = 1; Rst = 0;#100;
X = 0; Clk = 0; Rst = 0;#100;
X = 1; Clk = 1; Rst = 0;#100;
// Add stimulus here

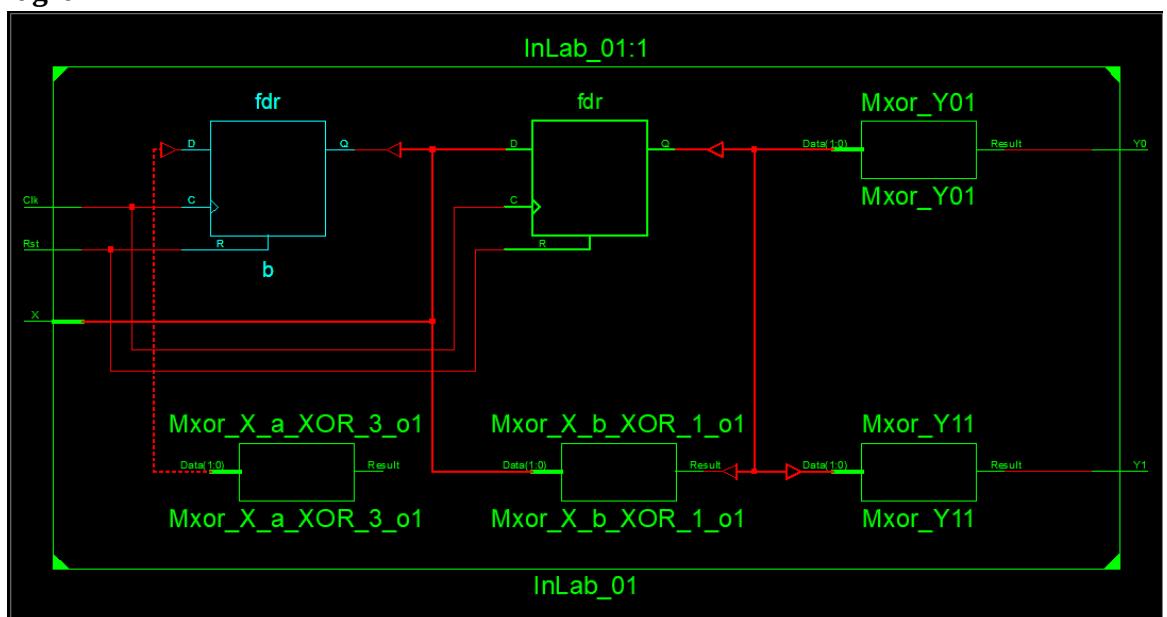
end

```

- Waveform

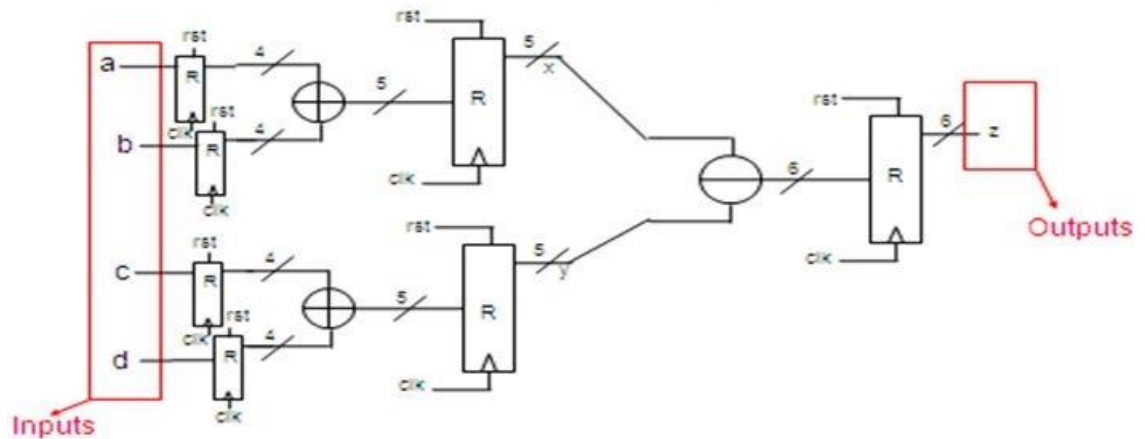


- RTL Diagram



1.2. Post Lab Task#1

Implement the Given Encoder in Verilog HDL



Solution:

- Verilog Code

```
module assign4(z, a, b, c, d, clk, rst);

//declaration of inputs, outputs, and wires
input wire [3:0]a, b, c, d;
input wire clk, rst;
output reg [4:0]z;
reg [4:0]a1, b1, c1, d1, a1b1, c1d1, fa1b1, fc1d1, xab;

//computational circuit code
always @(*)
begin
a1b1 = a1 ^ b1;
c1d1 = c1 ^ d1;
xab = fa1b1 ~^ fc1d1;
end

//sequential circuit code
//Synchronous clk and D Flip Flop
always @(posedge clk)
begin
if (rst)
begin
a1 <= 0;
b1 <= 0;
c1 <= 0;
d1 <= 0;
fa1b1 <= 0;
```

```

fc1d1 <= 0;
z <= 0;
end
else
begin
a1 <= a;
b1 <= b;
c1 <= c;
d1 <= d;
fa1b1 <= a1b1;
fc1d1 <= c1d1;
z <= xab;
end
end
endmodule

```

- **Verilog Test Bench**

```

`timescale 1ns / 1ps
module assign4TB;
    reg [3:0] a;
    reg [3:0] b;
    reg [3:0] c;
    reg [3:0] d;
    reg clk;
    reg rst;
    wire [4:0] z;
    assign4 uut (
        .z(z),
        .a(a),
        .b(b),
        .c(c),
        .d(d),
        .clk(clk),
        .rst(rst)
    );

    initial begin
        a = 4'b0000;
        b = 4'b0000;
        c = 4'b0000;
        d = 4'b0000;
        rst = 0;
        clk = 1;
    end
endmodule

```

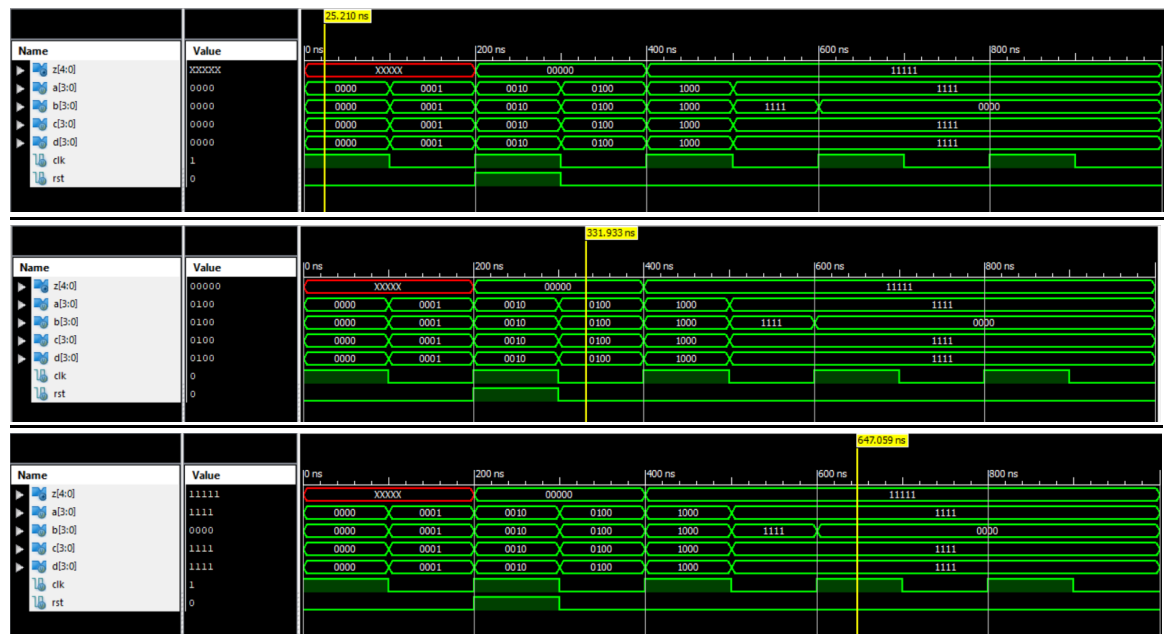
```

        #100;
        a = 4'b0001;
        b = 4'b0001;
        c = 4'b0001;
        d = 4'b0001;
        rst = 0;
        #100;
        a = 4'b0010;
        b = 4'b0010;
        c = 4'b0010;
        d = 4'b0010;
        rst = 1;
        #100;
        a = 4'b0100;
        b = 4'b0100;
        c = 4'b0100;
        d = 4'b0100;
        rst = 0;
        #100;
        a = 4'b1000;
        b = 4'b1000;
        c = 4'b1000;
        d = 4'b1000;
        rst = 0;
        #100;
        a = 4'b1111;
        b = 4'b1111;
        c = 4'b1111;
        d = 4'b1111;
        rst = 0;
        #100;
        a = 4'b1111;
        b = 4'b0000;
        c = 4'b1111;
        d = 4'b1111;
        rst = 0;

        #100;
    end
always #100 clk = ~clk;
endmodule

```

- Waveform



We have dry run while considering some values of inputs a, b, c, d.

Let's Suppose

a = 0000, b=0000, c=0000, d=0000,

the output will be Zero.

Let's Suppose

A = 1111, b=0000, c=1111, d=1111

The output will be 1111

We have cross verified this while doing dry run.

- RTL Diagram

